

CeylonPick

Software Requirements Specification (SRS)

IEEE 830 structure · Functional requirements by module · Business rules · NFRs · Interfaces · Traceability · Acceptance criteria

Version v1.0 · 6 September 2026 · Status: Baseline — the contract for the v1 build

Version History

Version	Date	Author	Summary
v1.0	6 September 2026	Founder (with AI-assisted drafting)	Initial baseline. Formalises the Feasibility Study, Project Plan, Technical Architecture, Marketing Plan, UI Specification and Page Overview into numbered, testable requirements.
v1.1	—	—	(reserved) After day-90 review: adjust business rules (COD caps, RTO thresholds) from measured data.
v2.0	—	—	(reserved) Campaign Board, reviews, referral, occasion pages, vendor self-onboarding.

Conventions

Requirement IDs are **FR-<MODULE>-nn** (functional), **NFR-nn** (non-functional), **BR-nn** (business rule), **IF-nn** (external interface). **Pri**: M = Must (v1 cannot ship without it), S = Should (v1 target, may slip to v1.1), C = Could (nice to have). **Rel**: release in which the requirement is delivered (v1 = 8-week build; v2 = months 4–6; v3 = months 7–12). *Shall* denotes a binding requirement. Every requirement has at least one acceptance criterion and a trace to a page (UI Spec) and/or module (Architecture).

Companion documents (normative where referenced): Technical Architecture v1.0 (state machine, ledger, modules) · UI Specification v1.0 (pages, components, states) · Project Plan v1.0 (KPIs, kill criteria) · Marketing Plan v1.0 · Page Overview v1.0.

1. Introduction

1.1 Purpose

This SRS specifies the functional and non-functional requirements of CeylonPick v1, a curated multi-vendor marketplace for verified Sri Lankan-made products with a creator (affiliate) storefront and commission system. It is the contract between the business plan and the build: a feature that is not here is not in v1.

1.2 Scope

In scope: public catalogue and creator storefronts; buyer checkout with Cash on Delivery (OTP-confirmed) and PayHere prepaid; order lifecycle to settlement; double-entry ledger with vendor and creator payouts; creator attribution; vendor and creator onboarding with manual verification; admin operations. Out of scope for v1: native mobile apps, microservices, Redis/message brokers, multi-currency and international shipping, vendor self-publishing without moderation, news/jobs/general content, clothing and electronics categories, automated bank payouts, buyer reviews (v2), Campaign Board (v2).

1.3 Definitions and acronyms

Term	Meaning
Buyer	Person placing an order; identified by mobile phone number, no password.
Creator / Affiliate	Approved content creator with a storefront page and attribution code who earns commission on settled orders.
Vendor / Maker	Manually verified Sri Lankan producer selling through the platform.
Admin	Founder/operator role with full access.
COD	Cash on Delivery — courier collects payment at the door.
OTP	One-time 6-digit code sent by SMS/WhatsApp to confirm a COD order or to log a buyer in.
RTO	Return to Origin — delivery failed or refused; parcel returned to vendor.
IPN	PayHere Instant Payment Notification — server-to-server webhook confirming payment.
Settlement	Order reaches SETTLED (delivered + 7 days, no open dispute); money is split in the ledger.
Held funds	Buyer money received but not yet split; never platform revenue.
Attribution	Linking an order to a creator via cookie (30 days, last click) or code at checkout.
SKU	A sellable product variant.
Kill switch	Admin setting that disables new orders while keeping tracking and dashboards available.

1.4 References

Technical Architecture v1.0 §6 (Order State Machine), §7 (Ledger), §8 (Attribution), §9 (Integrations) · UI Specification v1.0 §4–§9 · Project Plan v1.0 §3 (Business model), §8 (KPIs) · PayHere merchant integration documentation (checkout API, IPN) · Sri Lanka Personal Data Protection Act No. 9 of 2022 (for privacy obligations).

2. Overall Description

2.1 Product perspective

CeylonPick is a new, self-contained web system (Next.js front end, Spring Boot modular monolith, PostgreSQL) that integrates with PayHere (payments), one or more courier partners (delivery and COD remittance), a WhatsApp/SMS gateway (OTP and notifications) and S3-compatible object storage (images). It replaces no existing system; it competes with Facebook/Instagram selling for vendors and with Daraz for buyers.

2.2 Actors and permissions

Actor	Authentication	Can	Cannot
Visitor	None	Browse, search, view storefronts, apply as creator/vendor	Place order without phone
Buyer	Phone + OTP; 30-day session	Cart, checkout (COD/prepaid), track, dispute (≤48 h), reorder	See other buyers' data; cancel after CONFIRMED (must contact support)

Actor	Authentication	Can	Cannot
Creator	Email + password (+ optional OTP); approval required	Manage storefront, links, view attributed orders (masked), statements, bank details	See buyer identity; see other creators' data; change commission rates
Vendor	Email + password; verification required	Manage products (to moderation), pack/ship own orders, view own payables, request cancel	Publish unmoderated products; see other vendors; change order state except pack/ship
Admin	Email + password + OTP	Everything, with reason logging on state overrides, adjustments and settings	Edit or delete ledger entries; confirm payment without IPN
System (jobs)	Internal	OTP/payment timeouts, settlement, RTO-rate recompute, reconciliation, outbox dispatch	Skip guards of the state machine

2.3 Constraints

- Single developer; eight-week build timebox; infrastructure budget ≤ LKR 6,000/month (Architecture \$1).
- COD share expected 55–65% of orders; COD must be a first-class path with OTP confirmation.
- Card data is never handled by the system (PayHere hosted checkout) — PCI SAQ-A scope.
- Courier APIs may be unavailable at launch; manual status entry by admin/vendor must be fully supported.
- Three UI languages (EN/SI/TA) from v1 for buyer-facing pages.
- Registered business required for PayHere live account; sandbox for all development.

2.4 Assumptions and dependencies

- PayHere approves the merchant account before week 4 of the build; IPN is reliably delivered within seconds.
- At least one courier provides a CSV remittance report; at least one provides a status feed or the admin enters statuses manually.
- A WhatsApp Business API provider or local SMS gateway is available for OTP within 30 seconds of request.
- Vendors have smartphones with cameras and can upload packing photos.
- Legal entities: sole proprietorship at launch; vendor and creator terms accepted electronically with timestamp.

3. Business Rules

Business rules are configurable in Admin → Settings unless marked fixed. Changes are logged with actor and time and apply to orders placed after the change.

ID	Rule	Default (v1)	Configurable
BR-01	Platform commission on delivered order (percentage of product price)	6%	Yes (global)
BR-02	Creator commission, set per product by the vendor, within bounds	10%–20%	Bounds yes; value by vendor
BR-03	Payment gateway fee charged to vendor share on prepaid orders	Actual PayHere fee (~3%)	Yes
BR-04	Courier fee paid by buyer at checkout; platform never subsidises shipping	By district table	Fee table yes; rule fixed
BR-05	COD maximum order total (global) and per-category override	LKR 5,000; Beauty LKR 3,000	Yes
BR-06	Products flagged prepaid-only (e.g., personalised) cannot be ordered COD	Vendor flag	Per product
BR-07	COD orders require OTP confirmation; expiry 10 min per code; max 3 attempts; max 3 resends; order auto-cancels 24 h after placement if unconfirmed	As stated	Timings yes
BR-08	Prepaid orders auto-cancel if no successful IPN within 30 minutes	30 min	Yes
BR-09	Vendors must upload a packing photo before an order can be marked PACKED	Mandatory	Fixed
BR-10	Settlement occurs 7 days after DELIVERED if no open dispute	7 days	Yes
BR-11	Buyer dispute window after DELIVERED	48 hours	Yes
BR-12	Creator commission is recognised only at SETTLED; cancelled, unconfirmed, RTO, refunded orders earn nothing	Fixed	Fixed
BR-13	Vendor payable is recognised only at SETTLED	Fixed	Fixed
BR-14	Creator payouts on the 1st and 16th; minimum LKR 2,000; balances below roll forward	As stated	Yes
BR-15	Vendor payouts weekly on Friday for orders settled by Thursday 23:59	Weekly	Yes
BR-16	Creator RTO rate over trailing 30 days, with ≥10 attributed orders: >30% auto-pause; >40% flagged for removal	30% / 40% / 10	Yes
BR-17	Attribution: 30-day cookie, last click wins; code entered at checkout overrides cookie; frozen at PLACED	As stated	Cookie days yes
BR-18	Self-referral blocked: creator's registered phone may not be the buyer phone on an attributed order	Fixed	Fixed
BR-19	Single-vendor orders: a cart containing items from N vendors becomes N orders at checkout	Fixed (v1)	—
BR-20	Vendor cap per phase and SKU cap per vendor	25 vendors / 15 SKUs (phase 1)	Yes
BR-21	New or changed products require admin moderation before going live	Mandatory	Fixed
BR-22	Return-shipping fee on RTO charged to vendor per policy	LKR 250 flat (to confirm)	Yes
BR-23	Vendor-initiated cancellation after CONFIRMED requires admin approval and records a strike; 3 strikes → review	3	Yes
BR-24	Ledger entries are append-only; corrections only via ADJUSTMENT transactions with memo and admin identity	Fixed	Fixed
BR-25	Order state may change only via transitions in the state machine table (Architecture §6); admin overrides must target an allowed next state and record a reason	Fixed	Fixed

4. Functional Requirements

Grouped by module. Trace column references UI Specification pages (UI §n) and Architecture modules.

4.1 Authentication and accounts (AUTH)

ID	Requirement	Pri	Rel	Acceptance criteria	Trace
FR-AUTH-01	The system shall authenticate buyers by mobile phone number and a 6-digit OTP, without passwords.	M	v1	Buyer with valid SL mobile receives OTP ≤30 s; correct code creates 30-day session; no password field exists anywhere in buyer flows.	UI §5 My orders; orders/notifications
FR-AUTH-02	The system shall authenticate creators and vendors by email and password, with optional OTP second factor, and admins by email, password and mandatory OTP.	M	v1	Admin login without OTP is impossible; password policy ≥10 chars; 5 failed attempts → 15-min lockout.	web/auth
FR-AUTH-03	The system shall enforce role-based access so each actor can only reach pages and data listed in §2.2.	M	v1	Automated test: vendor A cannot read vendor B's orders via UI or API (403).	web/auth
FR-AUTH-04	Bank-detail edits by creators or vendors shall require OTP re-verification.	M	v1	Edit without fresh OTP is rejected.	UI §6/§7 Profile
FR-AUTH-05	The system shall record acceptance of creator/vendor terms with version and timestamp.	M	v1	Terms log visible in admin detail.	vendors/creators

4.2 Catalog (CAT)

ID	Requirement	Pri	Rel	Acceptance criteria	Trace
FR-CAT-01	The system shall present products by category with price filters and sort, showing verified badge, price, maker name and district on each card.	M	v1	Category page lists only LIVE products of VERIFIED vendors; filters are URL parameters.	UI §4 Category; catalog
FR-CAT-02	Each product shall have title, description, ≥1 image, price, category, variants with stock, COD-allowed flag, prepaid-only flag, lead time and maker story reference.	M	v1	Product cannot be published with missing mandatory fields.	catalog
FR-CAT-03	The product page shall show the delivery fee estimate and COD availability before the buy action.	M	v1	At 390 px width, fee/COD line is above the fold.	UI §4 Product
FR-CAT-04	The system shall reserve stock on order placement (15 min prepaid, 24 h COD) and release it on cancellation or timeout.	M	v1	Two concurrent orders for the last unit: exactly one succeeds.	orders/catalog
FR-CAT-05	The system shall provide keyword search over product titles, descriptions and vendor names.	S	v1	Search 'kithul' returns matching products and makers in <500 ms.	UI §4 Search
FR-CAT-06	The system shall support ISR/caching of category and product pages with a ≤10-minute staleness.	S	v1	Price change visible on public page within 10 min or on purge.	catalog; web
FR-CAT-07	The system shall enforce the vendor SKU cap (BR-20) with an explanatory message.	S	v1	16th product creation is refused with message.	vendors/catalog

4.3 Vendors (VEN)

ID	Requirement	Pri	Rel	Acceptance criteria	Trace
FR-VEN-01	Visitors shall be able to apply as a vendor with business/maker name, NIC or BR, product list, ≥3 photos, social links, district and phone.	M	v1	Application appears in admin queue within 1 min.	UI §4 Sell with us
FR-VEN-02	Admin shall verify a vendor by recording documents received, sample received and call completed before granting VERIFIED status.	M	v1	Products of unverified vendors are never publicly visible.	UI §8 Vendors

ID	Requirement	Pri	Rel	Acceptance criteria	Trace
FR-VEN-03	Vendors shall manage products, variants, stock and price, and set the creator commission within BR-02 bounds.	M	v1	Commission slider limits to 10–20%; stock edit is inline.	UI §7 Products
FR-VEN-04	Vendors shall see a 'Today' view with count of orders to pack, payable balance and next payout date.	M	v1	Numbers match orders in CONFIRMED and ledger VENDOR_PAYABLE balance.	UI §7 Today
FR-VEN-05	Vendors shall pack orders through a one-order-per-card flow requiring a packing photo before 'Mark packed' (BR-09).	M	v1	Mark packed disabled until photo uploaded; photo stored privately.	UI §7 Pack flow; orders
FR-VEN-06	Vendors shall enter courier tracking numbers to move an order to SHIPPED when no courier API exists.	M	v1	Tracking number validated non-empty; buyer notified.	orders
FR-VEN-07	Vendors shall view statements showing per-order price, creator commission, platform fee, gateway fee, RTO fees and net.	M	v1	Statement totals equal payout amount.	UI §7 Payouts; ledger
FR-VEN-08	Vendors shall request cancellation of a CONFIRMED order with a reason; admin approval applies BR-23.	S	v1	Strike count increments on approval.	orders
FR-VEN-09	Admin shall be able to suspend a vendor, hiding their products and pausing new orders while existing orders complete.	M	v1	Suspended vendor's products return 404/hidden within 10 min.	vendors

4.4 Creators (CRE)

ID	Requirement	Pri	Rel	Acceptance criteria	Trace
FR-CRE-01	Visitors shall apply as a creator with handle, platform, follower range, niche, language, phone and sample link.	M	v1	Application in admin queue; auto-reply promises 48 h.	UI §4 Earn
FR-CRE-02	Approved creators shall receive a public storefront at /@handle listing up to 24 picks with a personal note per pick.	M	v1	Storefront renders with OG image for link previews.	UI §4 Storefront
FR-CRE-03	Each creator shall have a unique code and shall be able to generate product-specific links carrying the code.	M	v1	Links resolve; clicks recorded with hashed IP and user agent.	creators
FR-CRE-04	Visiting a storefront or a link with a code shall set a 30-day attribution cookie; last click wins (BR-17).	M	v1	Two creators' links in sequence: second wins.	creators
FR-CRE-05	Buyers shall be able to enter a creator code at checkout; a valid code overrides the cookie (BR-17).	M	v1	Invalid code shows inline error; valid code shows creator name.	UI §5 Checkout
FR-CRE-06	The creator dashboard shall show payable balance, next payout date, 7-day funnel (clicks, confirmed, delivered, paid), RTO rate and recent orders with masked buyer identity.	M	v1	No buyer name/phone visible; numbers match ledger and orders.	UI §6 Dashboard
FR-CRE-07	The system shall recompute each creator's trailing-30-day RTO rate nightly and apply BR-16 auto-pause with notification.	M	v1	Creator at 31% with 12 orders is paused; new orders unattributed; admin sees trigger.	creators (job)
FR-CRE-08	Creators shall download payout statements listing settled orders and commissions.	M	v1	PDF/CSV totals equal payout.	UI §6 Payouts; ledger
FR-CRE-09	The system shall block self-referral per BR-18.	M	v1	Order with buyer phone = creator phone is stored without attribution and flagged.	orders/creators
FR-CRE-10	A Campaign Board shall let vendors post seeded campaigns and creators apply.	S	v2	—	UI §6 Campaign Board

4.5 Orders and fulfilment (ORD)

Order status shall follow the state machine in Architecture §6 exactly. The transitions below are requirements; the table there is normative.

ID	Requirement	Pri	Rel	Acceptance criteria	Trace
FR-ORD-01	Buyers shall place an order from a cart with phone, name, address and district; carts spanning N vendors shall create N orders (BR-19).	M	v1	Two-vendor cart → two order numbers, two shipments, one confirmation screen.	UI §5 Cart/Checkout; orders
FR-ORD-02	Buyers shall choose COD or prepaid; COD shall be hidden with explanation when total exceeds BR-05 or any item is prepaid-only (BR-06).	M	v1	LKR 5,200 hamper shows prepaid only with reason text.	UI §5 Checkout
FR-ORD-03	COD orders shall enter AWAITING_OTP and send an OTP; correct entry within limits moves to CONFIRMED; BR-07 limits and auto-cancel shall apply.	M	v1	3 wrong codes → CANCELLED with message; unconfirmed 24 h → CANCELLED by job; stock released.	orders/notifications
FR-ORD-04	Prepaid orders shall enter AWAITING_PAYMENT and move to CONFIRMED only on a verified PayHere IPN (never from the browser return).	M	v1	Return URL with success but no IPN → still pending; IPN → CONFIRMED.	payments/orders
FR-ORD-05	Vendors shall move CONFIRMED → PACKED (with photo) → SHIPPED (with tracking).	M	v1	Each transition recorded in status history with actor.	orders
FR-ORD-06	Courier status updates (webhook, poll or manual entry) shall move SHIPPED → DELIVERED or → RTO.	M	v1	Manual admin entry allowed with reason; buyer notified.	orders; courier adapter
FR-ORD-07	DELIVERED orders shall automatically move to SETTLED after BR-10 unless a dispute is open.	M	v1	Job runs hourly; order delivered at T settles at T+7d ±1 h.	orders (job)
FR-ORD-08	Buyers shall be able to open a dispute within BR-11 with reason and 1–3 photos, pausing settlement.	M	v1	Dispute at 47 h accepted; at 49 h refused with message.	UI §5 Dispute; orders
FR-ORD-09	Admin shall resolve disputes to SETTLED (rejected), REFUNDED (prepaid) or RTO (COD return), with templated messages to buyer and vendor.	M	v1	Each outcome posts the correct ledger transaction.	UI §8 Disputes
FR-ORD-10	Every state change shall be recorded in an order status history with from, to, actor, reason and timestamp.	M	v1	Tracking timeline and admin audit read from this history.	orders
FR-ORD-11	The system shall prevent any transition not in the state-machine table, including by admin (BR-25).	M	v1	API attempt SHIPPED → CONFIRMED returns 409 with allowed targets.	orders
FR-ORD-12	Buyers shall track an order by number via a link that does not require login, showing the timeline, packing photo, courier and amount due.	M	v1	WhatsApp link opens tracking page; no login prompt.	UI §5 Tracking
FR-ORD-13	The system shall generate printable order slips for a packing batch (order no., buyer name, phone, address, COD amount).	S	v1	Slip renders for N packed orders in one print view.	UI §7 Pack flow
FR-ORD-14	The system shall provide an admin kill switch disabling new orders while tracking and dashboards remain available.	M	v1	Toggle on → checkout shows 'Ordering paused' within 1 min.	UI §8 Settings

4.6 Payments (PAY)

ID	Requirement	Pri	Rel	Acceptance criteria	Trace
FR-PAY-01	The system shall integrate PayHere hosted checkout with server-side hash generation and shall never collect card data.	M	v1	Penetration checklist: no card fields in DOM; hash matches PayHere spec.	payments

ID	Requirement	Pri	Rel	Acceptance criteria	Trace
FR-PAY-02	The IPN endpoint shall verify signature, amount and currency, and shall be idempotent on PayHere payment_id.	M	v1	Replayed IPN → 200, no second state change, no second ledger txn.	payments
FR-PAY-03	Payment failure or 30-minute timeout shall cancel the order and release stock (BR-08).	M	v1	Job cancels at 30 min; buyer offered retry or COD.	payments/orders
FR-PAY-04	Refunds for prepaid disputes shall be recorded in the ledger and executed via PayHere/bank by admin with reference.	M	v1	REFUNDED requires refund reference field.	ledger/payments
FR-PAY-05	A daily reconciliation shall compare successful Payments to the PayHere settlement report and list exceptions.	S	v1	Missing/duplicate settlement appears in admin exceptions.	payments (job)

4.7 Ledger and payouts (LED)

The ledger shall implement Architecture §7 (accounts, balanced transactions, append-only).

ID	Requirement	Pri	Rel	Acceptance criteria	Trace
FR-LED-01	Every money movement shall be recorded as a balanced double-entry transaction; debits equal credits per transaction (enforced by the database).	M	v1	Unbalanced insert is rejected at commit.	ledger
FR-LED-02	Ledger entries shall be append-only; corrections shall be ADJUSTMENT transactions with memo and admin id (BR-24).	M	v1	No UPDATE/DELETE grants on ledger tables for the app role.	ledger
FR-LED-03	On DELIVERED (COD) the system shall post Dr COURIER_CLEARING / Cr HELD_FUNDS; on CONFIRMED (prepaid) Dr GATEWAY_CLEARING / Cr HELD_FUNDS.	M	v1	Worked example in Architecture Fig. 6 reproduces exactly.	ledger
FR-LED-04	On SETTLED the system shall split HELD_FUNDS into VENDOR_PAYABLE, CREATOR_PAYABLE, PLATFORM_REVENUE and GATEWAY_FEE per BR-01..03, idempotent per order.	M	v1	Re-running settlement job posts nothing new; rounding absorbed by vendor share (HALF_EVEN, 2 dp).	ledger
FR-LED-05	Admin shall import courier remittance CSVs, auto-match to shipments, manually match exceptions, and post REMITTANCE transactions.	M	v1	Unmatched rows remain listed until resolved.	UI §8 Remittance
FR-LED-06	The system shall show a reconciliation check: entered bank balance vs HELD_FUNDS + PLATFORM_REVENUE – payouts, with the difference and exceptions.	M	v1	Difference of 0 shows green; non-zero red with drill-down.	UI §8 Ledger
FR-LED-07	Admin shall run payouts per BR-14/15: preview payees above minimum → approve (locks, generates bank bulk CSV) → mark paid with references → PAYOUT transactions posted and payees notified.	M	v1	Approved run cannot be edited; CSV columns match bank template.	UI §8 Payouts
FR-LED-08	Creators and vendors shall see their payable balance, next payout date and statement history reflecting ledger state in near real time (≤1 min).	M	v1	Balance equals sum of payable account entries.	UI §6/§7
FR-LED-09	RTO fees (BR-22) shall be posted against vendor payable with order reference and shown on statements.	S	v1	RTO order shows negative line on vendor statement.	ledger

4.8 Notifications (NOT)

ID	Requirement	Pri	Rel	Acceptance criteria	Trace
FR-NOT-01	The system shall send OTP by SMS (primary) with WhatsApp as alternative, delivered within 30 s, rate-limited to 3 per phone per hour.	M	v1	4th request in an hour is refused with countdown.	notifications
FR-NOT-02	The system shall notify buyers on CONFIRMED, SHIPPED (with tracking link), DELIVERED (with dispute window) and CANCELLED/RTO, in the buyer's language.	M	v1	Each event produces exactly one message; logged with delivery status.	notifications
FR-NOT-03	The system shall notify vendors on new CONFIRMED orders, disputes, cancellations and payouts; creators on approval, pause, and payouts.	M	v1	Message log shows recipient and template.	notifications
FR-NOT-04	All notifications shall be event-driven via the outbox and idempotent per event id.	M	v1	Replayed event does not resend.	notifications; shared kernel
FR-NOT-05	Marketing messages to buyers shall require opt-in and be limited to one per week.	M	v1	Broadcast tool refuses second send in 7 days to same list.	notifications

4.9 Administration (ADM)

ID	Requirement	Pri	Rel	Acceptance criteria	Trace
FR-ADM-01	The admin dashboard shall show delivered orders vs weekly target, RTO 7d/30d, contribution margin per delivered order, payouts due, and actionable queues as counts that open filtered lists.	M	v1	All numbers reproducible from DB queries listed in the runbook.	UI §8 Ops
FR-ADM-02	Admin shall view any order with full history, evidence, IPN log and ledger entries, and transition it only to allowed next states with a mandatory reason (BR-25).	M	v1	Reason empty → transition refused.	UI §8 Order detail
FR-ADM-03	Admin shall moderate new/changed products before they go live (BR-21).	M	v1	Product stays DRAFT until approved.	UI §8 Products
FR-ADM-04	Admin shall manage settings for BR-01..BR-23 configurable items, message templates (EN/SI/TA), courier list, occasion banner and kill switch, with an audit log.	M	v1	Setting change shows actor/time; applies to new orders only.	UI §8 Settings
FR-ADM-05	Admin shall export orders, ledger transactions and payouts as CSV for accounting.	S	v1	Export matches on-screen totals for the same filter.	admin
FR-ADM-06	Admin shall approve/decline creator applications and pause/reactivate creators with reason.	M	v1	Paused creator's links stop attributing.	UI §8 Creators

4.10 Content and localisation (LOC)

ID	Requirement	Pri	Rel	Acceptance criteria	Trace
FR-LOC-01	All buyer-facing UI strings shall be available in English, Sinhala and Tamil, selectable in the header and persisted.	M	v1	Language switch changes all UI strings without reload errors; persists across sessions.	web
FR-LOC-02	Product content shall be stored in the vendor's language with optional translations; the UI shall show the best available.	S	v1	Missing translation falls back gracefully with no empty fields.	catalog
FR-LOC-03	Money shall display as 'LKR 12,450' and dates as 'Fri 16 Sep' in Asia/Colombo time everywhere.	M	v1	No raw decimals or ISO timestamps in buyer/creator/vendor UI.	web
FR-LOC-04	Static pages (How it works, About, Policies) shall exist in three languages at launch.	M	v1	All 3 language versions reachable.	UI §4

4.11 Release v2/v3 requirements (recorded for planning, not in v1 acceptance)

ID	Requirement	Pri	Rel	Acceptance criteria	Trace
FR-V2-01	Buyer reviews with photo on settled orders, moderated.	S	v2	—	UI v2
FR-V2-02	Buyer referral codes: both parties receive credit after the referred buyer's first delivered order.	S	v2	—	creators (owner BUYER)
FR-V2-03	Campaign Board with vendor-funded seeding and creator applications.	S	v2	—	campaigns
FR-V3-01	Occasion landing pages with curated SKUs, cut-off dates and creator briefs.	S	v3	—	UI v3
FR-V3-02	Signature collection with monthly placement fee posted as PLATFORM_REVENUE.	C	v3	—	ledger
FR-V3-03	Vendor self-onboarding to application stage with the same manual verification.	C	v3	—	vendors
FR-V3-04	Blog / gift guides (content marketing).	C	v3	—	web

5. Non-Functional Requirements

ID	Category	Requirement	Verification
NFR-01	Performance	Product page TTFB <300 ms from cache; checkout and dashboard APIs p95 <500 ms; LCP <2.5 s on a mid-range Android over 4G for Home and Product.	Lighthouse mobile ≥85 performance; k6 test at 100 concurrent users, 50 orders/hour.
NFR-02	Capacity (year 1)	5,000 daily visitors; 1,000 orders/day peak headroom on a single VPS; images ≤120 KB.	Load test; image pipeline check.
NFR-03	Availability	99.5% monthly for public and checkout; planned maintenance announced in-app.	Uptime monitor report.
NFR-04	Recovery	RPO 24 h (nightly pg_dump to R2), RTO 2 h (redploy image + restore).	Quarterly restore drill documented.
NFR-05	Security	HTTPS only; JWT in httpOnly SameSite cookies; CSRF protection; rate limiting per IP and phone; OWASP Top 10 checklist; dependency scanning in CI; secrets outside repo.	Checklist signed before go-live; automated scan in CI.
NFR-06	Data protection	No card data stored; bank details encrypted at rest (AES-GCM, key outside DB); IPs hashed; click data retained 90 days; buyer data deletable on request; privacy notice in EN/SI/TA (PDPA-aligned).	Code review; data-map document.
NFR-07	Auditability	Order status history, ledger append-only, admin actions and settings changes logged with actor and time.	Audit queries in runbook.
NFR-08	Money integrity	Ledger transactions always balanced (DB-enforced); settlement and payouts idempotent; bank reconciliation available on demand.	Unit + integration tests; reconciliation shows 0 on test data.
NFR-09	Usability	Mobile-first at 390 px; 44 px tap targets; 4.5:1 contrast; no colour-only meaning; every error names the fix; every list has an empty state.	UI Spec §9 checklist review; 5-user hallway test for checkout and pack flow.
NFR-10	Localisation	EN/SI/TA UI strings; native copy (not machine-translated) for buyer flows.	Native speaker review sign-off.
NFR-11	Maintainability	Modular monolith with enforced module boundaries (Spring Modulith verify in CI); Flyway migrations; ≥80% unit coverage on state machine and ledger.	CI report.
NFR-12	Observability	Errors to Sentry (FE/BE); health endpoint; daily digest email (orders by state, job lag, IPN failures, RTO).	Digest received daily.
NFR-13	Compatibility	Chrome/Android WebView, Safari iOS (last 2 major), desktop Chrome/Edge/Firefox; PWA installable.	Manual matrix before go-live.
NFR-14	Cost	Infrastructure ≤ LKR 6,000/month in year 1.	Monthly invoice review.

6. External Interface Requirements

ID	Interface	Requirement
IF-01	PayHere checkout	Server generates MD5 hash per PayHere spec; redirect to hosted checkout; sandbox/live keys separated by environment; amounts formatted to 2 dp exactly.
IF-02	PayHere IPN	POST /webhooks/payhere: verify md5sig, amount, currency; upsert Payment by payment_id; respond 200 within 2 s; processing inside a transaction; failures logged and retried by reconciliation.
IF-03	Courier port	CourierPort interface: createShipment(order) → trackingNo; fetchStatus(trackingNo); parseRemittance(csv). Adapters: ManualCourierAdapter (v1 mandatory), one API adapter when available. Status mapping table per courier to {IN_TRANSIT, DELIVERED, RETURNED, FAILED_ATTEMPT}.
IF-04	Messaging port	MessagingPort: send(template, to, vars). SMS adapter (local gateway) and WhatsApp Business API adapter (approved templates). Delivery status logged; retries with backoff; max 3 OTP per phone per hour.
IF-05	Object storage	S3-compatible (Cloudflare R2); browser uploads via pre-signed PUT; public product images via CDN; private packing/dispute photos via short-lived signed URLs.
IF-06	Bank payouts	Bulk-transfer CSV export matching the business bank's template; references captured manually; no bank API in v1.
IF-07	User interface	As specified in UI Specification v1.0: pages, components, states, status colours, languages.

ID	Interface	Requirement
IF-08	Analytics	Privacy-friendly page analytics (e.g., Plausible) plus internal attribution clicks; UTM conventions per Marketing Plan §12.2.

7. Data Requirements

Entity group	Key data	Retention / privacy
Buyer	Phone (identity), name, address, district, language, WhatsApp opt-in	Retained while orders exist; deletable on request (orders anonymised, ledger kept)
Vendor	Business/maker name, NIC/BR (document scan), district, pickup address, bank details (encrypted), commission agreement, strikes	Documents visible only to admin; deleted 12 months after account closure
Creator	Handle, platforms, follower range, niche, language, phone, bank details (encrypted), RTO stats	As vendor
Product / Variant	Title, description (per language), images, price, category, flags, stock, lead time, commission %	Indefinite; history via price snapshot on order items
Order / Item / Shipment	Snapshot prices and percentages, status, history, OTP challenge (hash only), payment, courier, tracking, packing photo key	Indefinite (financial record); photos 12 months
Attribution	Links, codes, clicks (hashed IP, UA, timestamp)	Clicks 90 days; aggregates indefinite
Ledger	Accounts, transactions, entries, payouts, remittance imports	Indefinite; append-only
Messages	Template, recipient, status	90 days
Audit	Status history, admin actions, settings changes	Indefinite

Identifiers are ULIDs; money is NUMERIC(12,2) LKR; timestamps UTC stored, Asia/Colombo displayed. Full schema: Technical Architecture §5 and §7.

8. Traceability Matrix

Requirement group	UI pages (UI Spec)	Modules (Architecture)	Business rules	Plan KPI served
AUTH	Login/OTP screens, Profiles	web/auth, notifications	—	Repeat rate (frictionless login)
CAT	Home, Category, Product, Search, Vendor Products, Admin Products	catalog	BR-05, 06, 20, 21	Delivered orders (conversion)
VEN	Sell with us, Vendor Today/Pack/Orders/Products/Payouts, Admin Vendors	vendors, orders, ledger	BR-09, 20–23	Verified vendors; RTO
CRE	Earn, Storefront, Creator dashboard/links/picks/orders/payouts, Admin Creators	creators, ledger	BR-02, 12, 16–18	Active paid creators; attributed share
ORD	Cart, Checkout, OTP, Payment result, Tracking, Dispute, Pack flow, Admin Orders/Disputes	orders, payments, notifications	BR-05–11, 19, 25	Delivered orders; RTO
PAY	Checkout, Payment result, Admin Ledger exceptions	payments	BR-03, 08	Prepaid share
LED	Creator/Vendor payouts, Admin Ledger/Payouts/Remittance	ledger	BR-01–03, 10, 12–15, 22, 24	Contribution margin; payout on time
NOT	(all flows)	notifications, shared kernel	BR-07	Confirmation rate; repeat rate
ADM	Admin area	all	BR-25 and configurable rules	Weekly review; kill criteria
LOC	All public/buyer pages	web, catalog	—	Reach outside English-speaking Colombo

9. Acceptance Test Scenarios (end-to-end)

v1 is accepted when all scenarios pass on the live environment with sandbox payments and at least one real courier delivery.

ID	Scenario	Expected outcome
AT-01	COD happy path: creator link → product → checkout COD → OTP → vendor packs with photo → tracking entered → admin marks delivered → 7 days → settled → payouts run	Order passes PLACED→...→SETTLED; ledger shows Fig. 6 postings; creator and vendor statements show correct amounts; buyer received 4 messages.
AT-02	Prepaid happy path with PayHere sandbox and IPN	CONFIRMED only after IPN; GATEWAY_FEE posted; return URL alone never confirms.
AT-03	OTP failure: 3 wrong codes	Order CANCELLED; stock released; creator sees 'unconfirmed', no commission; buyer message explains.
AT-04	OTP timeout: no code for 24 h	Job cancels; same effects as AT-03.
AT-05	RTO: courier returns parcel	Order RTO; return fee posted to vendor; creator RTO rate increments; no commission.
AT-06	Dispute within 48 h with photos; admin refunds (prepaid) / returns (COD)	Settlement paused; correct terminal state and ledger; both parties notified.
AT-07	Dispute after 48 h	Refused with message; settlement proceeds.
AT-08	Two-vendor cart	Two orders, two shipments, one confirmation; each settles independently.
AT-09	COD above cap / prepaid-only item	COD option hidden with explanation.
AT-10	Duplicate IPN replay	No second confirmation, no duplicate ledger transaction.
AT-11	Settlement job re-run	No duplicate SETTLEMENT transactions (idempotent).
AT-12	Creator auto-pause	Creator with 4 RTO of 12 orders paused overnight; links stop attributing; admin sees reason.
AT-13	Self-referral	Order with buyer phone = creator phone stored unattributed and flagged.
AT-14	Remittance import with one unmatched row	Matched rows post; unmatched listed; reconciliation shows difference until resolved.
AT-15	Payout run	Preview → approve → CSV → mark paid; balances zeroed; notifications sent; run immutable.
AT-16	Invalid admin transition	API/UI refuses; allowed targets shown; reason mandatory on valid override.
AT-17	Kill switch	New orders blocked within 1 min; tracking and dashboards work.
AT-18	Localisation	Full COD flow completed in Sinhala and Tamil UI without English fallbacks in buyer-facing strings.
AT-19	Performance	Lighthouse mobile ≥85 on Home and Product; k6 100 concurrent users without errors.
AT-20	Security	Vendor A cannot access vendor B data (403); admin login requires OTP; no card fields present.

10. Open Issues

#	Issue	Owner	Needed by
1	Confirm courier partner(s) and whether a status API/webhook exists (affects FR-ORD-06, IF-03).	Founder	Build week 2
2	Confirm SMS/WhatsApp provider and template approval lead time (FR-NOT-01/02).	Founder	Build week 2
3	Decide RTO fee policy value (BR-22) — flat LKR 250 vs 50/50 split.	Founder	Build week 5
4	Bank bulk-transfer CSV template (IF-06).	Founder	Build week 5
5	PayHere live merchant approval timing (FR-PAY-01).	Founder	Build week 4
6	Native Sinhala/Tamil copywriting resource for buyer flows (NFR-10).	Founder	Build week 7

End of document — v1.0